Interpretable Deep Learning Algorithms for Prokaryotic Genome Annotation

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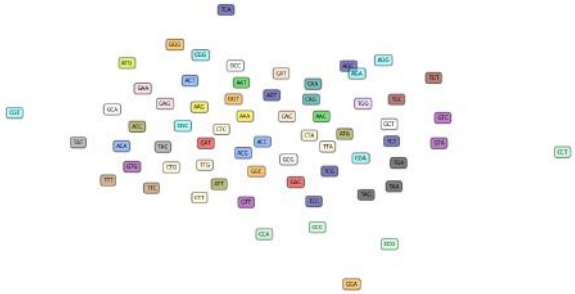
Problems With Genome Annotation

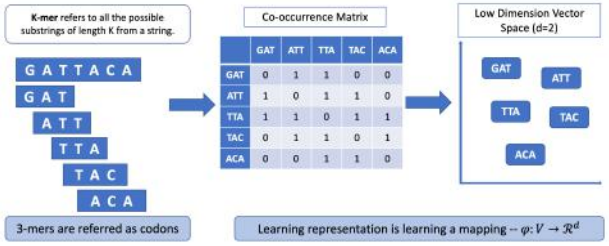
- Current Genome Annotation Problems:
  - Imprecise annotations:
    - 12 thousand NCBI bacterial genomes have wrong rRNA ends!
  - Detailed annotations not present in canonical files:
    - 5'UTRs/3'UTRs, transcription factor binding sites, etc.
- Idea: Deep network to scan over all genomic sequences
  - Multi-class classification for all types of annotations
  - Integrate detailed information from Model Organism Databases

Word Embeddings – 2D

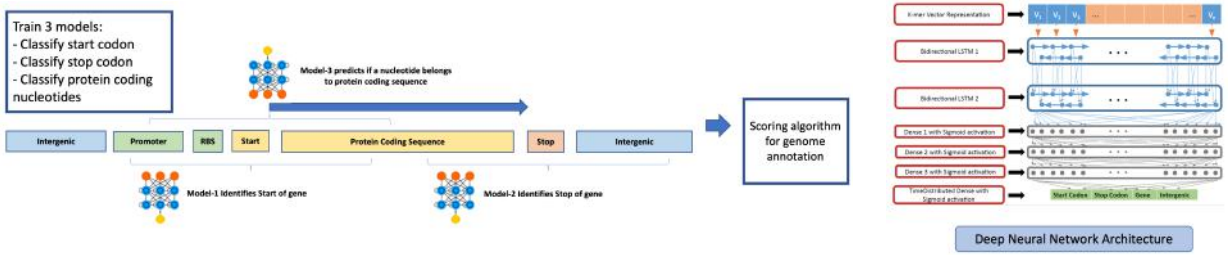
- Trained embeddings on 2B bases of bacterial genomes
- Used 3-mer words, with 1-shift
- Project weight vectors onto 2D (TSNE)
  - 64 3-mers, i.e. codons
- Synonymous Codons Show Similarity
  - No clusters, but synonymous codons collocate



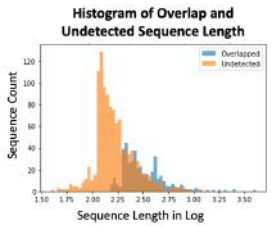
Data Representation: K-mer Embeddings

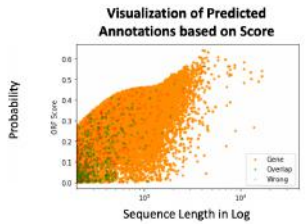


Workflow of Deep Annotator Algorithm with Model Architecture



Evaluating Deep Annotator for Prokaryotic Genome Annotation





Deep Annotator Vs Popular Annotation Algorithm

Method	Precision (%)	Recall (%)	F-score (%)
GeneMarkS	92.78	96.68	94.69
Prodigal	95.22	95.87	95.55
Glimmer	94.08	92.30	93.18
DeepAnn	94.95	93.23	94.08

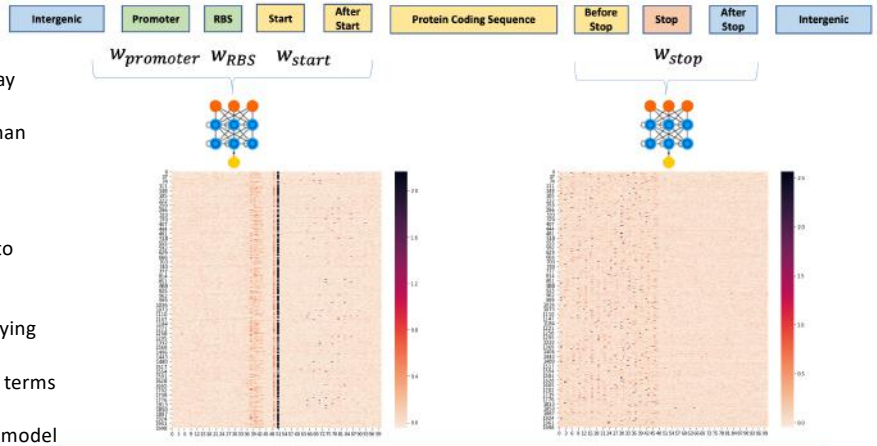
Interpretable Deep Learning Model for Prokaryotic Genome Annotation

Observations:

- We are sequencing billions of nucleotides (biological data) every day
- Understanding these data and discovering features are beyond human capability

Challenges:

- Deep learning model should be able to handle high cardinality heterogenous features
- The model should capture the underlying mechanism of biological significance
- The model should be interpretable in terms of its learned representations
- We should be able to reuse a trained model weights in a second related task



Hitmap showing the weights of genomic features resulting in better prediction